CSW-332 MACHINE LEARNING AND APPLICATIONS

Time Allotted = 90 min.

Answer all questions –

1. Write a Python program for the Decision Tree (DT) supervised machine learning classifier using the Iris dataset. The program should demonstrate the following operations:
    a. Load the Iris Dataset and split it into training and testing samples.
    b. Train the DT ML classifier.
    c. Evaluate the model using performance metrix derived from Confusion Matrix.
    d. Validate the classifier using multiple test samples.
    e. Visualize the Decision Tree.

2. Write a Python program for k-means Clustering in Machine Learning using the Wine dataset. The program should demonstrate the following operations:
    a. Load the Wine Dataset and perform feature scaling.
    b. Train the k-means Clustering ML classifier.
    c. Reduce dimensions for visualization using Principal Component Analysis (PCA).
    d. Plot the clusters and the centroids.
    e. Evaluate the model using performance metrix derived from Confusion Matrix.

3. Write a Python program for Logistic Regression in Machine Learning using the Iris dataset. The program should demonstrate the following operations:
    a. Load the Iris dataset and create a Data Frame.
    b. Select the independent variable and the dependent variable.
    c. Split the dataset into training and testing samples.
    d. Train the Logistic Regression model.
    e. Evaluate the model using performance metrix derived from Confusion Matrix.
    f. Predict the output using test data.